

Draft of Econ 303 Lecture Notes. Part 1.

Bo E. Honoré

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Abstract

This is the first draft of my lecture notes for the first part of the course. Please be aware that I plan update them during the semester. For the lastes version, see <http://www.princeton.edu/~econ303>.

1 Ordinary Least Squares.

We have n observations of the form (y_i, x_i) for $i = 1, 2, \dots, n$. Would like to model the y 's as an exact function of the x 's. Impossible.

We can:

- Approximate y as well as possible by some function of x . (A linear function, for example).
- Think of (y_i, x_i) as the realization of a random vector (Y_i, X_i) . In that case, we might want to estimate $h(x_i) = E[Y_i | X_i = x_i]$. This is hard without additional assumptions. But what if we assumed that it is linear?
- Think of (y_i, x_i) as the realization of a random vector (Y_i, X_i) . Suppose we just want to have a model that predicts Y_i as well as possible by a linear function of X_i , then we might want to try to find parameter that make $E[(Y_i - (b_1 + b_2 X_i))^2]$ small.

It will turn out that these three ways to think about the problem will lead to the same calculations, but the interpretation of the results will differ.

1.1 Curve Fitting.

Let's try to get the best possible fit

$$\begin{aligned}
 \min Q(b_1, b_2) &= \sum_{i=1}^n (y_i - b_1 - b_2 x_i)^2 \\
 &= \sum_{i=1}^n y_i^2 + b_1^2 + b_2^2 x_i^2 - 2y_i b_1 - 2y_i b_2 x_i + 2b_1 b_2 x_i \\
 &= nb_1^2 + b_1 \sum_{i=1}^n (2b_2 x_i - 2y_i) + \text{constant} \\
 &= b_2^2 \sum_{i=1}^n x_i^2 + b_2 \sum_{i=1}^n (2b_1 x_i - 2x_i y_i) + \text{constant}
 \end{aligned}$$

Proposition 1 *The function*

$$c_1 z^2 + c_2 z + c_3, \quad (c_1 > 0)$$

is minimized at

$$2c_1 z + c_2 = 0 \quad \blacksquare$$

Let's apply this to minimization of $Q(b_1, b_2)$ with respect to b_1 :

$$2nb_1 + 2 \sum_{i=1}^n b_2 x_i - 2 \sum_{i=1}^n y_i = 0$$

or

$$-2 \sum_{i=1}^n (y_i - b_1 - b_2 x_i) = 0 \tag{1}$$

Now minimize $Q(b_1, b_2)$ with respect to b_2 :

$$2b_2 \sum_{i=1}^n x_i^2 + \sum_{i=1}^n (2b_1 x_i - 2x_i y_i) = 0$$

or

$$-2 \sum_{i=1}^n x_i (y_i - b_1 - b_2 x_i) = 0 \tag{2}$$

(1) and (2) are called the normal equations. Solving them gives

$$\hat{\beta}_2 = \frac{\sum_{i=1}^n (x_i - \bar{x}) (y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{\sum_{i=1}^n (x_i - \bar{x}) y_i}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{\sum_{i=1}^n (x_i - \bar{x}) y_i}{\sum_{i=1}^n (x_i - \bar{x}) x_i} = \sum_{i=1}^n w_i y_i$$

and

$$\hat{\beta}_1 = \bar{y} - \hat{\beta}_2 \bar{x} = \sum_{i=1}^n \left(\frac{1}{n} - \bar{x} w_i \right) y_i$$

where

$$w_i = \frac{(x_i - \bar{x})}{\sum_{j=1}^n (x_j - \bar{x}) x_j} = \frac{(x_i - \bar{x})}{\sum_{j=1}^n (x_j - \bar{x})^2}$$

Let

$$e_i = y_i - \hat{y}_i$$

where

$$\hat{y}_i = \hat{\beta}_1 + \hat{\beta}_2 x_i$$

then

$$(1) \implies \sum_{i=1}^n e_i = 0$$

$$(2) \implies \sum_{i=1}^n e_i x_i = 0$$

Note that

$$\begin{aligned} TSS &= \sum_{i=1}^n (y_i - \bar{y})^2 \\ &= \sum_{i=1}^n ((y_i - \hat{y}_i) + (\hat{y}_i - \bar{y}))^2 \\ &= \sum_{i=1}^n e_i^2 + \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 + 2 \sum_{i=1}^n e_i (\hat{y}_i - \bar{y}) \\ &= ? \end{aligned}$$

To make progress note that

$$\begin{aligned}\hat{y}_i - \bar{y} &= \hat{\beta}_1 + \hat{\beta}_2 x_i - \bar{y} \\ &= \left(\bar{y} - \hat{\beta}_2 \bar{x} \right) + \hat{\beta}_2 x_i - \bar{y} \\ &= \hat{\beta}_2 (x_i - \bar{x})\end{aligned}$$

so

$$TSS = \sum_{i=1}^n e_i^2 + \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 + 2 \sum_{i=1}^n e_i \hat{\beta}_2 (x_i - \bar{x}) = ?$$

Next note that

$$2 \sum_{i=1}^n e_i \hat{\beta}_2 (x_i - \bar{x}) = 2 \hat{\beta}_2 \sum_{i=1}^n e_i x_i - 2 \hat{\beta}_2 \bar{x} \sum_{i=1}^n e_i = 0$$

hence

$$\begin{aligned}TSS &= \sum_{i=1}^n e_i^2 + \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 \\ &= RSS + ESS\end{aligned}$$

Define

$$R^2 = \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2} = 1 - \frac{\sum_{i=1}^n e_i^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

then

$$0 \leq R^2 \leq 1$$

R^2 can be used to measure the goodness of fit.

2 Statistical Properties of OLS estimator.

Consider the model

$$Y_i = \beta_1 + \beta_2 x_i + u_i$$

Assumptions Assume

1. x_i nonstochastic
2. $E[u_i] = 0$

3. $V[u_i] = \sigma^2$
4. $cov(u_i, u_j) = 0$ for $i \neq j$.
5. $u_i \sim N(0, \sigma^2)$

In this section we will write Y_i to denote the random variable, and y_i to denote the realized value of the dependent variable.

Proposition 2 *Make assumptions 1-2. Then*

$$\begin{aligned} E[\hat{\beta}_1] &= \beta_1 \\ E[\hat{\beta}_2] &= \beta_2 \end{aligned}$$

Under assumptions 1-4,

$$\begin{aligned} V[\hat{\beta}_1] &= \sigma^2 \frac{\sum_{i=1}^n x_i^2}{n \sum_{i=1}^n (x_i - \bar{x})^2} \\ V[\hat{\beta}_2] &= \sigma^2 \frac{1}{\sum_{i=1}^n (x_i - \bar{x})^2} \end{aligned}$$

Proof. : First note that for any constant c ,

$$\sum_{i=1}^n w_i c = 0.$$

Second, note that

$$E[Y_i] = E[\beta_1 + \beta_2 x_i + u_i] = \beta_1 + \beta_2 x_i$$

and that therefore

$$E[\bar{Y}] = \beta_1 + \beta_2 \bar{x}$$

Applying these to the equations for $\hat{\beta}_1$ and $\hat{\beta}_2$ gives

$$\begin{aligned} E[\hat{\beta}_2] &= E\left[\sum_{i=1}^n w_i y_i\right] = \sum_{i=1}^n w_i E[y_i] \\ &= \sum_{i=1}^n w_i (\beta_1 + \beta_2 x_i) = \beta_2 \sum_{i=1}^n w_i x_i \\ &= \beta_2 \frac{\sum_{i=1}^n (x_i - \bar{x}) x_i}{\sum_{j=1}^n (x_j - \bar{x}) x_j} = \beta_2 \end{aligned}$$

and

$$\begin{aligned} E \left[\widehat{\beta}_1 \right] &= E \left[\bar{Y} - \widehat{\beta}_2 \bar{x} \right] = E \left[\bar{Y} \right] - E \left[\widehat{\beta}_2 \right] \bar{x} \\ &= \beta_1 + \beta_2 \bar{x} - \beta_2 \bar{x} = \beta_1 \end{aligned}$$

To calculate the variances, note that

$$V \left[Y_i \right] = V \left[\beta_1 + \beta_2 x_i + u_i \right] = V \left[u_i \right] = \sigma^2$$

so

$$\begin{aligned} V \left[b_2 \right] &= V \left[\sum_{i=1}^n w_i y_i \right] = \sum_{i=1}^n w_i^2 V \left[y_i \right] \\ &= \sigma^2 \sum_{i=1}^n w_i^2 = \sigma^2 \sum_{i=1}^n \left(\frac{(x_i - \bar{x})}{\sum_{j=1}^n (x_j - \bar{x})^2} \right)^2 \\ &= \sigma^2 \sum_{i=1}^n \frac{(x_i - \bar{x})^2}{\left(\sum_{j=1}^n (x_j - \bar{x})^2 \right)^2} \\ &= \sigma^2 \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{\left(\sum_{j=1}^n (x_j - \bar{x})^2 \right)^2} \\ &= \sigma^2 \frac{1}{\sum_{j=1}^n (x_j - \bar{x})^2} \end{aligned}$$

To calculate $V \left[\widehat{\beta}_1 \right]$ is it useful to note $\sum_{j=1}^n (x_j - \bar{x})^2 = \left(\sum_{j=1}^n x_j^2 \right) - n\bar{x}^2$

$$\begin{aligned} V \left[\widehat{\beta}_1 \right] &= \sum_{i=1}^n \left(\frac{1}{n} - \bar{x} w_i \right)^2 \sigma^2 \\ &= \sum_{i=1}^n \left(\frac{1}{n^2} - 2\bar{x} w_i \frac{1}{n} + \bar{x}^2 w_i^2 \right)^2 \sigma^2 \\ &= \left(\frac{1}{n} - 0 + \bar{x}^2 \sum_{i=1}^n w_i^2 \right) \sigma^2 \\ &= \left(\frac{1}{n} + \frac{\bar{x}^2}{\sum_{j=1}^n (x_j - \bar{x})^2} \right) \sigma^2 \end{aligned}$$

$$\begin{aligned}
&= \left(\frac{\sum_{j=1}^n (x_j - \bar{x})^2}{n \sum_{j=1}^n (x_j - \bar{x})^2} + \frac{n\bar{x}^2}{n \sum_{j=1}^n (x_j - \bar{x})^2} \right) \sigma^2 \\
&= \left(\frac{\left(\sum_{j=1}^n x_j^2 \right) - n\bar{x}^2}{n \sum_{j=1}^n (x_j - \bar{x})^2} + \frac{n\bar{x}^2}{n \sum_{j=1}^n (x_j - \bar{x})^2} \right) \sigma^2 \\
&= \frac{\sigma^2 \sum_{j=1}^n x_j^2}{n \sum_{j=1}^n (x_j - \bar{x})^2}
\end{aligned}$$

■

Let

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^n e_i^2}{n-2}$$

Proposition 3 $\hat{\sigma}^2$ is an unbiased estimator of σ^2 :

$$E[\hat{\sigma}^2] = \sigma^2$$

Proof. No proof will be given here. ■

Proposition 4 (Gauss Markov Theorem) *Make assumption 1-4. The ordinary least squares estimators, $\hat{\beta}_1$ and $\hat{\beta}_2$ have the lowest variances of all estimators that (a) are unbiased estimators of β_1 and β_2 , and (b) linear function of $Y_1 \dots Y_n$.*

Proof. No proof will be given here. ■

Proposition 5 *Under assumption 1-5*

$$\begin{aligned}
\hat{\beta}_1 &\sim N(\beta_1, \sigma_{b_1}^2) \\
\hat{\beta}_2 &\sim N(\beta_2, \sigma_{b_2}^2)
\end{aligned}$$

or

$$\frac{\hat{\beta}_k - \beta_k}{\sigma_{b_k}} \sim N(0, 1), \quad k = 1, 2$$

Proof. Note that

$$Y_i \sim N(\beta_1 + \beta_2 x_i, \sigma^2)$$

The first two results then follow from the fact that $\hat{\beta}_1$ and $\hat{\beta}_2$ are linear combinations of independent normals, and linear combinations of independent normals is normal. The third results follows from the first two. ■

Let

$$\begin{aligned}\hat{\sigma}_{b_1}^2 &= \hat{\sigma}^2 \frac{\sum_{i=1}^n x_i^2}{n \sum_{i=1}^n (x_i - \bar{x})^2} \\ \hat{\sigma}_{b_2}^2 &= \hat{\sigma}^2 \frac{1}{\sum_{i=1}^n (x_i - \bar{x})^2}\end{aligned}$$

Proposition 6 *Under assumptions 1–5*

$$\frac{\hat{\beta}_k - \beta_k}{\hat{\sigma}_{b_k}} \sim T_{n-2}, \quad k = 1, 2. \quad (3)$$

Proof. The result will not be proved here. ■

Remark 1 *Equation (3) implies that 95% confidence intervals for β_1 and β_2 can be constructed by*

$$\hat{\beta}_k \pm t_{n-2, 0.975} \cdot \hat{\sigma}_{b_k},$$

where $t_{n-2, 0.975}$ is a number that you look up in a t -distribution.

Remark 2 *In large samples, the Central Limit Theorem can be used to justify these results, even if the errors are not normally distributed. In large samples, it also does not matter (much) whether you use a normal or a T -distribution.*

3 Multivariate Regression.

We now turn to some complications of the situations. First, what if we want to have more than one explanatory variable:

$$Y_i = \beta_1 + X_{2i}\beta_2 + \dots + X_{ki}\beta_k + u_i$$

Note that we have changed the notation. From now on, we will use capital letters Y and X to denote the dependent variable and the explanatory variables, respectively. We will use the lower case, y and x , to deviations from averages, so

$$\begin{aligned}y_i &= Y_i - \bar{Y} \\ x_{ji} &= X_{ji} - \bar{X}_j\end{aligned}$$

Again, it makes sense to minimize the sum of squared errors

$$\sum_{i=1}^n (Y_i - (b_1 + X_{2i}b_2 + \dots + X_{ki}b_k))^2$$

It is not too difficult to show that the $(b_1, b_2, \dots, b_k)$ that minimizes the sum of squared errors (call it $(\hat{\beta}_1, \hat{\beta}_2, \dots, \hat{\beta}_k)$) must satisfy the equations

$$\sum_{i=1}^n \left(Y_i - \left(\hat{\beta}_1 + X_{2i}\hat{\beta}_2 + \dots + X_{ki}\hat{\beta}_k \right) \right) = 0 \quad (4)$$

$$\sum_{i=1}^n \left(Y_i - \left(\hat{\beta}_1 + X_{2i}\hat{\beta}_2 + \dots + X_{ki}\hat{\beta}_k \right) \right) X_{2i} = 0 \quad (5)$$

$$\vdots \quad (6)$$

$$\sum_{i=1}^n \left(Y_i - \left(\hat{\beta}_1 + X_{2i}\hat{\beta}_2 + \dots + X_{ki}\hat{\beta}_k \right) \right) X_{ki} = 0 \quad (7)$$

These can be written as

$$\begin{aligned} \sum_{i=1}^n e_i &= 0 \\ \sum_{i=1}^n e_i X_{2i} &= 0 \\ &\vdots \\ \sum_{i=1}^n e_i X_{ki} &= 0 \end{aligned}$$

where the residual e_i is defined by

$$e_i = Y_i - \left(\hat{\beta}_1 + X_{2i}\hat{\beta}_2 + \dots + X_{ki}\hat{\beta}_k \right)$$

Solving these k equations with k unknowns $(\hat{\beta}_1, \hat{\beta}_2, \dots, \hat{\beta}_k)$ is in principle no problem. Unfortunately, the notation is a real mess, so we will not write down the general expressions here.

However, it is instructive to look some of the expressions for the case where $k = 3$. In that case

$$\hat{\beta}_1 = \bar{Y} - \hat{\beta}_2 \bar{X}_2 - \hat{\beta}_3 \bar{X}_3$$

$$\hat{\beta}_2 = \frac{(\sum_{i=1}^n y_i x_{2i}) (\sum_{i=1}^n x_{3i}^2) - (\sum_{i=1}^n y_i x_{3i}) (\sum_{i=1}^n x_{2i} x_{3i})}{(\sum_{i=1}^n x_{2i}^2) (\sum_{i=1}^n x_{3i}^2) - (\sum_{i=1}^n x_{2i} x_{3i})^2}$$

$$\hat{\beta}_3 = \frac{(\sum_{i=1}^n y_i x_{3i}) (\sum_{i=1}^n x_{2i}^2) - (\sum_{i=1}^n y_i x_{2i}) (\sum_{i=1}^n x_{2i} x_{3i})}{(\sum_{i=1}^n x_{2i}^2) (\sum_{i=1}^n x_{3i}^2) - (\sum_{i=1}^n x_{2i} x_{3i})^2}$$

$$E [\hat{\beta}_1] = \beta_1$$

$$E [\hat{\beta}_2] = \beta_2$$

$$E [\hat{\beta}_3] = \beta_3$$

$$V [\hat{\beta}_1] = \left[\frac{1}{n} + \frac{\bar{X}_2^2 \sum_{i=1}^n x_{3i}^2 + \bar{X}_3^2 \sum_{i=1}^n x_{2i}^2 - 2 \bar{X}_2 \bar{X}_3 \sum_{i=1}^n x_{2i} x_{3i}}{(\sum_{i=1}^n x_{2i}^2) (\sum_{i=1}^n x_{3i}^2) - (\sum_{i=1}^n x_{2i} x_{3i})^2} \right] \cdot \sigma^2$$

$$V [\hat{\beta}_2] = \frac{\sum_{i=1}^n x_{3i}^2}{(\sum_{i=1}^n x_{2i}^2) (\sum_{i=1}^n x_{3i}^2) - (\sum_{i=1}^n x_{2i} x_{3i})^2} \cdot \sigma^2$$

$$V [\hat{\beta}_3] = \frac{\sum_{i=1}^n x_{2i}^2}{(\sum_{i=1}^n x_{2i}^2) (\sum_{i=1}^n x_{3i}^2) - (\sum_{i=1}^n x_{2i} x_{3i})^2} \cdot \sigma^2$$

To gain some intuition, let

$$S_j^x = \frac{1}{n-1} \sum_{i=1}^n (X_{ji} - \bar{X}_j)^2 = \frac{1}{n-1} \sum_{i=1}^n x_{ji}^2$$

denote the sample variance of X_j and let

$$S_{j\ell}^x = \frac{1}{n-1} \sum_{i=1}^n (X_{ji} - \bar{X}_j) (X_{\ell i} - \bar{X}_\ell) = \frac{1}{n-1} \sum_{i=1}^n x_{ji} x_{\ell i}$$

denote the sample covariance between X_j and X_ℓ .

We can then write

$$\begin{aligned}
V[\hat{\beta}_3] &= \frac{\sum_{i=1}^n x_{2i}^2}{(\sum_{i=1}^n x_{2i}^2)(\sum_{i=1}^n x_{3i}^2) - (\sum_{i=1}^n x_{2i}x_{3i})^2} \cdot \sigma^2 \\
&= \frac{\frac{1}{n-1} \sum_{i=1}^n x_{2i}^2}{\left(\frac{1}{n-1} \sum_{i=1}^n x_{2i}^2\right) \left(\frac{1}{n-1} \sum_{i=1}^n x_{3i}^2\right) - \left(\frac{1}{n-1} \sum_{i=1}^n x_{2i}x_{3i}\right)^2} \cdot \frac{1}{n-1} \cdot \sigma^2 \\
&= \frac{S_2^x}{S_2^x S_3^x - (S_{23}^x)^2} \cdot \frac{1}{n-1} \cdot \sigma^2 \\
&= \frac{1}{1 - \left(\frac{S_{23}^x}{\sqrt{S_2^x S_3^x}}\right)^2} \cdot \frac{1}{S_3^x} \cdot \frac{1}{n-1} \cdot \sigma^2 \\
&= \frac{1}{1 - (r_{23}^x)^2} \cdot \frac{1}{S_3^x} \cdot \frac{1}{n-1} \cdot \sigma^2
\end{aligned}$$

where r_{23}^x is the sample correlation between X_j and X_ℓ . The last expression for $V[\hat{\beta}_3]$ above have a nice interpretation, because it states that $V[\hat{\beta}_3]$ is big if

- r_{23}^x is close to either 1 or -1 . This makes intuitive sense because if r_{23}^x is close to 1 then X_2 will tend to be big exactly when X_3 is big and small exactly when X_3 is small, and it will therefore be very difficult to disentangle the effect of X_2 on Y from the effect of X_3 on Y . A similar story can be told when r_{23}^x is close to -1 .
- S_3^x is close to 0. This makes sense because if S_3^x is close to 0, then X_3 is close to a constant, and it will be difficult for distinguish the effect from X_3 from the constant.
- n is small. This makes sense because we expect that we can make more precise statements with more observations.
- σ^2 is large. If you think of u_i as measurement error this is obvious. Big measurement errors will lead to imprecision.

3.1 Multicollinearity

The case where two explanatory variables are highly correlated is a special case of what is called called *multicollinearity*. In general multicollinearity

means that one of the explanatory variables is highly correlated with some linear combination of the rest. To see why this can cause problems, consider the linear regression model

$$Y_i = \beta_1 + X_{2i}\beta_2 + \dots + X_{k-1,i}\beta_{k-1} + X_{ki}\beta_k + u_i \quad (8)$$

Assume that you can write X_{ki} as an exact linear combination of the other explanatory variables,

$$X_{ki} = a_1 + X_{2i}a_2 + \dots + X_{k-1,i}a_{k-1}$$

One might refer to this as exact multicollinearity. For any constant, c , we can then write

$$X_{ki}c = (a_1 + X_{2i}a_2 + \dots + X_{k-1,i}a_{k-1})c$$

and hence

$$X_{ki}\beta_k = X_{ki}\beta_k(1 - c) + (a_1 + X_{2i}a_2 + \dots + X_{k-1,i}a_{k-1})c$$

so we can write (8) as

$$\begin{aligned} Y_i &= \beta_1 + X_{2i}\beta_2 + \dots + X_{k-1,i}\beta_{k-1} + X_{ki}\beta_k + u_i \\ &= \beta_1 + X_{2i}\beta_2 + \dots + X_{k-1,i}\beta_{k-1} + X_{ki}\beta_k(1 - c) \\ &\quad + (a_1 + X_{2i}a_2 + \dots + X_{k-1,i}a_{k-1})c + u_i \\ &= (\beta_1 + a_1c) + X_{2i}(\beta_2 + a_2c) + \\ &\quad \dots + X_{k-1,i}(\beta_{k-1} + a_{k-1}c) + X_{ki}\beta_k(1 - c) + u_i \end{aligned}$$

so there is no way that you can ever tell the difference between whether the true parameters are

$$\beta_1, \beta_2, \dots, \beta_{k-1}, \beta_k$$

or

$$(\beta_1 + a_1c), (\beta_2 + a_2c), \dots, (\beta_{k-1} + a_{k-1}c), \beta_k(1 - c)$$

(because they imply the same relationship between Y_i and $X_{2i}, \dots, X_{k-1,i}, X_{ki}$).

The intuition behind the calculation is quite simple. If you can write X_{ki} as an exact linear combination of the other explanatory variables, then there is no way of knowing whether Y_i depends on X_{ki} or on some combination of the other explanatory variables.

If X_{ki} is close to (but not exactly equal to) a linear combination of the other explanatory variables, then the calculation isn't valid, but it is still intuitively clear that we should not expect to be able to precisely distinguish between whether Y_i depends on X_{ki} or on some combination of the other explanatory variables. The expression for $V[b_3]$ above confirms this intuition.

3.2 Estimation of σ^2 .

We estimate $V[u_i] = \sigma^2$ by

$$\hat{\sigma}^2 = \frac{1}{n-k} \sum_{i=1}^n e_i^2$$

It can be proved that this is an unbiased estimator of σ^2 .

3.3 Some statistical properties.

We will make the same assumptions as we did when we just had one regressor:

Assumptions. Assume

1. x nonstochastic
2. $E[u_i] = 0$
3. $V[u_i] = \sigma^2$
4. $\text{cov}(u_i, u_j) = 0$ for $i \neq j$.
5. $u_i \sim N(0, \sigma^2)$

Proposition 7 (Gauss Markov Theorem) *Make assumption 1-4. The ordinary least squares estimators, $\hat{\beta}_1, \hat{\beta}_2, \dots, \hat{\beta}_k$ have the lowest variances of all estimators that (a) are unbiased estimators of $\beta_1, \beta_2, \dots, \beta_k$, and (b) linear function of $Y_1 \dots Y_n$.*

Proposition 8 *Under assumption 1-5*

$$\frac{\hat{\beta}_j - \beta_j}{\hat{\sigma}_{b_j}} \sim T_{n-k}, \quad k = 1, \dots, k. \quad (9)$$

Proof. The result will not be proved here. ■

Remark 3 *Equation (3) implies that 95% confidence intervals for β_1 and β_2 can be constructed by*

$$\hat{\beta}_k \pm t_{n-k, 0.975} \cdot \hat{\sigma}_{b_k},$$

where $t_{n-k, 0.975}$ is a number that you look up in a t -distribution with $n - k$ degrees of freedom. In large samples, the Central Limit Theorem can be used to justify these results, even if the errors are not normally distributed. In large samples, it also does not matter (much) whether you use a normal or a T -distribution.

3.4 F-tests

In the linear regression model, suppose we want to test

$$\beta_{j_1} = \overline{B}_{j_1}, \quad \beta_{j_2} = \overline{B}_{j_2}, \dots \quad \beta_{j_p} = \overline{B}_{j_p}$$

simultaneously. The alternative is that at least one of them is different.

Idea: If null is true then it shouldn't matter much whether we estimate the original model or one in which we have imposed the restrictions that $\beta_{j_1} = \overline{B}_{j_1}, \beta_{j_2} = \overline{B}_{j_2}, \dots, \beta_{j_p} = \overline{B}_{j_p}$. On the other hand, if the restrictions are not true then imposing them is "bad" and we would be better off estimating the parameters rather than imposing their values. So suppose that we estimate a restricted and an unrestricted model. Then

- The R^2 's in the two regressions should be close. If the restrictions are not true, then the restricted model should give a much worse fit.
- The sum of squared errors ($\sum e_i^2$) in the two regressions should be close. If the restrictions are not true, then the restricted model should give bigger errors (and hence bigger sum of squared errors).
- When we estimate the unrestricted model, $\hat{\beta}_{j_1}$ should be close to $\overline{B}_{j_1}$, $\hat{\beta}_{j_2}$ should be close to $\overline{B}_{j_2}, \dots$ and $\hat{\beta}_{j_p}$ should be close to $\overline{B}_{j_p}$.

All of these insights can be used to motivate the F -test, which is defined by

$$\begin{aligned} F &= \frac{(\sum (e_i^*)^2 - \sum e_i^2) / p}{\sum e_i^2 / (n - k)} \\ &= \frac{(R^2 - R_*^2) / p}{(1 - R^2) / (n - k)} \\ &= \text{something that has to do with } \hat{\beta}_{j_1} - \overline{B}_{j_1}, \hat{\beta}_{j_2} - \overline{B}_{j_2}, \dots, \hat{\beta}_{j_p} - \overline{B}_{j_p} \end{aligned}$$

Where n is the number of observations, k is the number of parameters in the unrestricted model, p is the number of restrictions to be imposed, and $*$ denotes quantities associated with the restricted model.

Clearly large values of F are evidence against the null. It turns out that the distribution of F can be calculated. It depends on $n - k$ and p and is

denoted an F -distribution with $n - k$ and p degrees of freedom. Tables exist for F -distributions.

It is easy to do F -tests in STATA (see example “<http://www.princeton.edu/~econ303/ex3.html>”)

Suppose that we wanted to test whether none (not including the constant) of the explanatory variables help explain the dependent variable. The R^2 in a model that has only a constant is 0 (why?), so the F -test is

$$F = \frac{(R^2) / (k - 1)}{(1 - R^2) / (n - k)}$$

This F -test statistic is automatically reported by Stata.

3.5 More about R^2

The interpretation of R^2 is that it measures the fraction of the variability of the dependent variable that can be explained by the explanatory variables. One might therefore use R^2 to compare two sets of explanatory variables. The F -tests above make this formal in one context.

R^2 is often used as an informal “quick and dirty” way to decide how good a job the regressors do in explaining the dependent variable. Clearly adding more explanatory variables cannot decrease R^2 (why?). If the aim is to get a high R^2 one should therefore throw in everything in sight as explanatory variables. Clearly, this does not make sense, and it is therefore often more interesting to consider the *adjusted* R^2 :

$$\overline{R}^2 = 1 - (1 - R^2) \frac{n - 1}{n - k}$$

The $\overline{R}^2$ adjusts R^2 downwards as the number of explanatory variables increases.

Note that it does not make sense to compare the R^2 for models with different dependent variables.

3.6 Dummy Variables.

4 Non-normality in the Linear Regression Model.

So far, we have maintained the assumption that $\varepsilon \sim N(0, \sigma^2 I)$. We will now investigate what happens (and what we can do about it) if this assumption

is not satisfied. First, we consider the case where $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_n$ are i.i.d. with mean 0 and variance σ^2 , but not necessarily normally distributed. If you review your notes from earlier, it will be clear that many of the results that we obtained, did not rely on normality. Let me remind you of some of the properties that hold even if the ε 's are not normal:

- (a) $\hat{\beta}$ is an unbiased estimator of β .
- (b) $V[\hat{\beta}] = \sigma^2(X'X)^{-1}$.
- (c) $\hat{\beta}$ is BLUE (best linear unbiased estimator).
- (d) s^2 is an unbiased estimator of σ^2 .
- (e) $\hat{\beta}$ and e are uncorrelated (if there is a constant in the regression).

However, it is not true that $\hat{\beta}$ is normal, that $\hat{\beta}$ is independent of s^2 , that the t-statistics are t-distributed, etc. (in other words all the results concerning distributions go down the drain).

Hence, we must rely on *asymptotic theory*. This is a pretty advanced subject. Essentially the key results are that

- averages of random variables get close to their expectation. This is known as the Law of Large Numbers.
- averages of random variables are approximately normally distributed (after they have been normalized appropriately). This is known as the Central Limit Theorem.

With those two facts, one can show that under appropriate assumptions, the OLS estimator is approximately normally distributed. This is in itself of some interest, but the most important implication is that the test-statistics and confidence intervals that we constructed before, are still approximately valid even if the normality assumption is not satisfied (provided that the other assumptions are satisfied).

The following are examples of (simple) Laws of Large Numbers and Central Limit Theorems. They are provided to give you the flavor of the types of results you use

Proposition 9 (Law of Large Numbers) *Let $X_1, X_2, X_3, \dots$ be a sequence of independent and identically distributed random variables with $\mu = E[X_i]$. Then for any $\varepsilon > 0$,*

$$P\left(\mu - \varepsilon < \frac{1}{n} \sum_{i=1}^n X_i < \mu + \varepsilon\right) \longrightarrow 1 \quad \text{as} \quad n \longrightarrow \infty.$$

So the Law of Large Numbers says that $\frac{1}{n} \sum_{i=1}^n X_i$ gets arbitrarily close to μ (with very high probability) as n gets big.

Proposition 10 (Central Limit Theorem) *Let $X_1, X_2, X_3, \dots$ be a sequence of independent and identically distributed random variables with $\mu = E[X_i]$ and $\sigma^2 = V[X_i]$. Let $Z_n = \sqrt{n} \left(\frac{1}{n} \sum_{i=1}^n X_i - \mu \right) / \sigma$. Then for any a*

$$P(Z_n \leq a) \longrightarrow \Phi(a) \quad \text{as} \quad n \longrightarrow \infty.$$

Note that if $X_1, X_2, X_3, \dots$ are normally distributed then

$$Z_n = \sqrt{n} \left(\frac{1}{n} \sum_{i=1}^n X_i - \mu \right) / \sigma$$

has a standard normal distribution, so for any a , $P(Z_n \leq a) = \Phi(a)$. The Central Limit Theorem says that that is true even if $X_1, X_2, X_3, \dots$ are not normally distributed.

5 Heteroskedasticity and Serial Correlation.

Consider the linear regression model

$$Y_i = \beta_1 + \beta_2 X_{2i} + \beta_3 X_{3i} + \dots + \beta_k X_{ki} + u_i$$

where we are interested in estimating the parameters $\beta_1, \beta_2, \beta_3, \dots, \beta_k$.

We earlier assumed that $V[u_i] = \sigma^2$ (i.e., the variance is the same for all observations) and that $\text{cov}(u_i, u_j) = 0$ for $i \neq j$. As pointed out in class, the consequences of this not being true are

- (a) that the proof of the Gauss-Markov theorem falls apart, and
- (b) that the expressions for the variance of the OLS estimator become incorrect.

Failure of the Gauss-Markov theorem means that OLS is no longer the best linear unbiased estimator. One way to think about this is that the confidence intervals are slightly larger than they could have been if we had used a different estimator.

The expressions for the variance of the OLS estimator being incorrect is much more important. The implication is that the confidence intervals are *incorrect*, as are the t-tests and F-tests that we calculate.

This raises two questions:

- how do we detect heteroskedasticity and serial correlation,
- what can we do about it if we conclude that heteroskedasticity or serial correlation is a problem.

Unfortunately, it turns out that the answers to these questions depend on whether serial correlation or heteroskedasticity is the problem.

5.1 Detecting heteroskedasticity.

Recall that we have assumed that $E[u_i] = 0$. This implies that $E[u_i^2] = V[u_i]$. If u_i is homoskedastic then its variance should be the same for all i .

Suppose one ran the regression

$$u_i^2 = G_1 + G_2 Z_{2i} + G_3 Z_{3i} + \dots + G_m Z_{mi} + \text{error} \quad (10)$$

where $Z_{2i}, Z_{3i}, \dots, Z_{mi}$ are some explanatory variables for individual i . Under homoskedasticity, the mean of u_i^2 should be the same for all observations. So one should not be able to explain the level of u_i^2 by $Z_{2i}, Z_{3i}, \dots, Z_{mi}$. Hence $G_2, G_3, \dots, G_m$ in (10) should all be 0. One could therefore test whether u_i is homoskedastic by running the regression in (10) and then test whether $G_2, G_3, \dots, G_m$ are all equal to 0. You could do this by looking at the F-test which STATA prints out automatically.

Of course, we do not know u_i . In (10) we therefore replace u_i by the residuals defined by

$$e_i = \left(Y_i - \hat{\beta}_1 + \hat{\beta}_2 X_{2i} + \hat{\beta}_3 X_{3i} + \dots + \hat{\beta}_k X_{ki} \right)$$

The resulting test is a version of a general class of tests called LM-tests. In some books, you will see a test based on the sample size times R^2 in the regression (10). Specifically, it can be shown that with no heteroskedasticity,

$$nR^2 \sim \chi_q^2$$

So you can perform the same test by calculating nR^2 , and you would reject homoskedasticity if it is too big to have come from a χ_q^2 -distribution.

5.2 What can you do if you detect heteroskedasticity?

As pointed out earlier, the biggest problem with heteroskedasticity is that the standard errors, confidence intervals and t - and F -tests are all incorrect. It turns out that one can construct standard errors, confidence intervals and t - and F -tests that are correct¹ even under heteroskedasticity (see expression 11.2.2 in Gujarati). The theory behind this is beyond the scope of this course. Implementing it in STATA is, however, trivial. All you have to do is put a “, robust” at the end of your “reg” command.

When people use this procedure, they often refer to it as calculating heteroskedasticity-consistent standard errors, White² standard errors, Eicker-White standard errors or Huber standard errors (the “h” in “hreg” stands for Huber).

GLS: See Gujarati 11.3

5.3 Serial Correlation.

Often in econometric work, the observations that we use to estimate our regression models correspond to consecutive time periods (in this case it is usually more intuitive to denote an observation by a subscript of t rather than i). Frequently, when the data consist of time series, the assumption that the error terms for different observations are uncorrelated will be violated: positive residuals tend to be followed by positive residuals and negative residuals by negative residuals. It seems likely that in such cases $\text{cov}(u_t, u_s) \neq 0$ (at least for some t and s). This phenomenon is called serial correlation or autocorrelation. From our previous discussion, we know that OLS estimation is unbiased. However, OLS will be inefficient and the standard errors produced by regression packages may be seriously biased. The most common assumption is that the u_t follow a first order autoregressive (AR1) model. That is, that

$$u_t = \rho u_{t-1} + v_t$$

¹Actually the procedure is only approximately correct, where the approximation can be expected to be good in large samples. You may ask why one would not *always* use the White standard errors. The answer is that they are approximate. If one knew that heteroskedasticity was not a problem, then the regular standard errors might be more precise.

²Hal White was the valedictorian here in the early 70's. An annual prize is given in his honor to an outstanding Economics major.

where the v_t are i.i.d. with mean zero and some variance σ^2 and $|\rho| < 1$. (A second order autoregressive model would look like $u_t = \rho_1 u_{t-1} + \rho_2 u_{t-2} + v_t$.)

If we knew ρ then we could eliminate the effect of the serial correlation as follows. We have³

$$Y_t = \beta_1 + \beta_2 X_t + u_t$$

and

$$Y_{t-1} = \beta_1 + \beta_2 X_{t-1} + u_{t-1}.$$

So that

$$Y_t - \rho Y_{t-1} = \beta_1(1 - \rho) + \beta_2(X_t - \rho X_{t-1}) + \rho u_t - \rho u_{t-1}$$

or

$$Y_t - \rho Y_{t-1} = \beta_1(1 - \rho) + \beta_2(X_t - \rho X_{t-1}) + v_t$$

Since v_t will have the properties of a good error term, estimating the above model by OLS will be appropriate.

In practice ρ will be unknown. We therefore apply the *Cochrane-Orcutt* procedure:

Step 1: Estimate $Y_t = \beta_1 + \beta_2 X_t + u_t$ by OLS and get the residuals e_t .

Step 2: Estimate $e_t = \rho e_{t-1} + v_t$ by OLS to get an estimate $\hat{\rho}$.

Step 3: Transform the data as above using $\hat{\rho}$ instead of ρ .

Step 4: Estimate the transformed data model by OLS.

Step 5: (optional) Iterate until the process converges.

Computational comment: STATA will perform the Cochrane-Orcutt procedure if you replace "reg" by "corc".

5.4 Detecting Serial Correlation.

The most frequently used measure of serial correlation is the DURBIN-WATSON statistics. It is defined by

$$DW = \frac{\sum_{t=2}^T (e_t - e_{t-1})^2}{\sum_{t=2}^T e_t^2}$$

It is a fact that $0 \leq DW \leq 4$, and that $DW \simeq 2(1 - \hat{\rho})$, where $\hat{\rho}$ is an estimate of the correlation between ε_t and ε_{t-1} . Hence values of DW far

³To simplify the notation, we assume that there is only one explanatory variable.

away from 2 are critical for the hypothesis of no serial correlation. Exact critical values cannot be found but bounds are reported in many text-books.

Computational Comment: STATA will calculate DW for you if you use “regdw” in place of “reg”.

Another test is the LM-test described next. The LM-test is really a test of the hypothesis that u_t is not correlated with $u_{t-1}, u_{t-2}, u_{t-3}, \dots, u_{t-q}$ (q is some number that you choose). Consider the regression

$$e_t = G_1 + G_2 X_t + \sum_{p=1}^q e_{t-p} \rho_p + v_t \quad (11)$$

We test for no serial correlation by testing that $\rho_1 \dots \rho_q$ are all equal to 0. The presence of $G_1 + G_2 X_t$ in (11) is difficult to explain (why it is needed, is an advance topic covered in graduate econometrics courses). In some books, you will see a test based on the sample size times R^2 in the regression (11). Specifically, it can be shown that with no serial correlation,

$$nR^2 \sim \chi_q^2$$

So you can perform the same test by calculating nR^2 , and you would reject that there is no serial correlation if it is too big to have come from a χ_q^2 -distribution.

The LM-test is considered superior to the Durbin Watson test because it is valid under much more general conditions. For example, the Durbin Watson is not correct if one of the explanatory variables is last periods y .

6 Notation

From now on, we will use the notation $x'\beta$ to denote $x_1\beta_1 + x_2\beta_2 + x_3\beta_3 + \dots + x_k\beta_k$. We will also use $x'_i\beta$ to denote $x_{i1}\beta_1 + x_{i2}\beta_2 + x_{i3}\beta_3 + \dots + x_{ik}\beta_k$.

We should have started this a long time ago!

We will also start thinking of the explanatory variables as realizations of random variables. This will have some consequences, but mostly, it will mean that we write $E[y|x]$ rather than $E[y]$.

7 Simple Binary Choice Models.

It is often the case that the variable of interest in a regression model is discrete/qualitative. Examples:

- To join a union or not.
- To work or not.
- Which brand to choose.
- Which mode of transportation to choose.

Consider first the case where there are two possible outcomes. Denote them by $y = 0$ or $y = 1$. There are a number of ways to specify such models.

The *Linear Probability Model* is the easiest to think about but often clearly inadequate:

$$y = x'\beta + \varepsilon, \quad E[\varepsilon|x] = 0$$

or

$$E[y|x] = x'\beta,$$

so the model can be estimated by OLS.

Note that y is binomial (conditional on x) so

$$x'\beta = E[y|x] = P(y = 1|x), \quad V[\varepsilon|x] = V[y|x] = P(y = 1|x) - P(y = 1|x)^2.$$

So the errors are heteroskedastic. We could do OLS and correct the standard errors, or we could do GLS (note that I have not really discussed GLS in these notes)

The problem with the linear probability model is that we need $0 \leq x'_i\beta \leq 1$ for the model to make sense. Even if $0 \leq x'_i\hat{\beta} \leq 1$, for all i , predictions could cause problems.

We will therefore turn to alternative models.

In general there are three ways to think about the problem:

- “Reduced” model $P(y_i = 1|x_i)$ as some function of an index $x'_i\beta$, so $P(y_i = 1|x_i) = F(x'_i\beta)$, for some fixed F and β is the parameter of interest.
- Structural “threshold-crossing”:

$$y_i^* = x'_i\beta - \varepsilon_i, \quad y_i = \begin{cases} 1 & \text{if } y_i^* \geq 0 \\ 0 & \text{if } y_i^* < 0 \end{cases}$$

Then $P(y_i = 1|x_i) = F(x_i\beta)$ where F is the cdf for ε . You can think of y_i^* as being the desirability of taking some action, $y_i^* = 1$, and the action being taken if that desirability is sufficiently high.

- Random utility model. Think of choice of mode of transportation. Assume two possible modes with indirect utilities:

$$\begin{aligned} U_{i0} &= \alpha_0 + z'_{i0}\beta + w'_i\gamma_0 + \varepsilon_0 \\ U_{i1} &= \alpha_1 + z'_{i1}\beta + w'_i\gamma_1 + \varepsilon_1 \end{aligned}$$

where z_{i0} and z_{i1} are mode characteristics (time, price, etc.) and w_i are socioeconomic characteristics (such as income and marital status). Then

$$\begin{aligned} P(\text{mode } 0) &= P(U_{i1} \leq U_{i0}) = P(\alpha_1 + z'_{i1}\beta + w'_i\gamma_1 + \varepsilon_1 \leq \alpha_0 + z'_{i0}\beta + w'_i\gamma_0 + \varepsilon_0) \\ &= P((\varepsilon_1 - \varepsilon_0) \leq (\alpha_0 - \alpha_1) + (z'_{i0} - z'_{i1})\beta + w'_i(\gamma_0 - \gamma_1)) \\ &= F_{\varepsilon_1 - \varepsilon_0}((\alpha_0 - \alpha_1) + (z'_{i0} - z'_{i1})\beta + w'_i(\gamma_0 - \gamma_1)) \end{aligned}$$

The advantage of this approach is that it is in principle straightforward to generalize it to more than two options. Moreover, the random utility formulation may yield natural interpretations of the parameters.

It should be clear that from the point of view of estimation, the three models are identical, but the interpretations are different.

Common choices for F :

- $F(\eta) = \begin{cases} 1 & \text{for } 1 \leq \eta \\ \eta & \text{for } 0 \leq \eta \leq 1 \\ 0 & \text{for } \eta \leq 0 \end{cases} \quad (\text{linear probability model})$
- $F(\eta) = \Phi(\eta) \quad (\text{Probit})$
- $F(\eta) = \frac{e^\eta}{1+e^\eta} \quad (\text{Logit}).$

The Logit and Probit look alike but logistic distribution has variance $\pi^2/3$.

7.1 Estimation.

Before proceeding by discussion various methods for estimation of the binary choice models, it is important to note that we must impose location and scale normalizations. To see this, consider the threshold crossing model with

$\varepsilon_i \sim N(\mu, \sigma^2)$ and $\beta = (\alpha_1, \beta_1)$ (α_1 is the intercept and β_1 the slopes). Let $\tilde{x}_i$ denote the explanatory variables, not including the constant. Then

$$\begin{aligned} P(y_i = 1|x_i) &= P(\varepsilon_i \leq \alpha_1 + \tilde{x}_i' \beta_1 | x_i) \\ &= P\left(\frac{\varepsilon_i - \mu}{\sigma} \leq \frac{\alpha_1 - \mu + \tilde{x}_i' \beta_1}{\sigma} \middle| x_i\right) = \Phi\left(\frac{\alpha_1 - \mu + \tilde{x}_i' \beta_1}{\sigma}\right) \end{aligned}$$

so we could never tell the difference between this model and one that has $\varepsilon_i \sim N(0, 1)$ and $\beta = ((\alpha_1 - \mu)/\sigma, \beta_1/\sigma)$.

- **Maximum Likelihood.** Likelihood function:

$$\mathcal{L} = \prod_i F(x_i' \beta)^{y_i} (1 - F(x_i' \beta))^{1-y_i}$$

and

$$\ln \mathcal{L} = \sum_i y_i \ln F(x_i' \beta) + (1 - y_i) \ln(1 - F(x_i' \beta))$$

7.1.1 Relationship between Probit, Logit and Linear Probability Model.

For the discussion in this section, it is easiest to think about the threshold-crossing model. In the notation for that model, ε_i is uniformly distributed (on the interval between 0 and 1) for the linear probability model, normally distributed (with mean 0 and variance 1) for the probit model, and logistically distributed for the logit model. As those three distributions have different variances, one would expect the scales for the estimated β 's to be different. Also, since the uniform does not have mean zero, one would expect the intercepts for that model to differ from those of the logit and probit (even after everything has been re-scaled). The following has been found to yield a good approximation if the probabilities are in the range between 0.3 and 0.7:

$$\begin{aligned} \hat{\beta}_{OLS} &\approx 0.4 \cdot \hat{\beta}_{probit} \quad \text{for the slopes, and} \quad \hat{\beta}_{OLS} \approx 0.4 \cdot \hat{\beta}_{probit} + 0.5 \quad \text{for the intercept} \\ \hat{\beta}_{OLS} &\approx 0.25 \cdot \hat{\beta}_{logit} \quad \text{for the slopes, and} \quad \hat{\beta}_{OLS} \approx 0.25 \cdot \hat{\beta}_{logit} + 0.5 \quad \text{for the intercept} \end{aligned}$$

However, it is important to note that the uniform, the normal and the logistic distributions differ in their shape as well as in their mean and variance. This means that the Probit estimators will be inconsistent (even if they are translated as above) if the true distribution of ε_i is not normal (even though the inconsistencies are small in practice). The same applies to the other distributions.

8 Maximum Likelihood.

This is confusing, so let us consider a really simple example. You flip a coin 5 times. You assume that each flip is independent and that the probability of getting a head is p . Suppose you observe head, head, tail, tail, head. You could then ask yourself: *What is the probability of that?* The answer would be

$$p \cdot p \cdot (1 - p) \cdot (1 - p) \cdot p = p^3 \cdot (1 - p)^2$$

Now, since you did observe head, head, tail, tail, head, it ought to be the case that the value of p is such that $p^3 \cdot (1 - p)^2$ is pretty big (because otherwise, you probably would not have observed head, head, tail, tail, head). It therefore (?!?) follows that we might want to estimate p by the value that maximizes $p^3 \cdot (1 - p)^2$.

If you took the right classes, *and* if you remember what they were about, *then* you will know that the value of p that maximizes $p^3 \cdot (1 - p)^2$ is the same as the value that maximizes

$$\ln(p^3 \cdot (1 - p)^2) = 3 \ln(p) + 2 \ln(1 - p)$$

Now, if you still remember those classes, then you will know that the value of p that maximizes $3 \ln(p) + 2 \ln(1 - p)$ is $\frac{3}{5}$. This is an example of a maximum likelihood estimator. Here is the general setup....

You have a number of observations with joint density function (or probability distribution function) f . Suppose that f depends on some unknown parameter θ . The maximum likelihood estimator maximizes the joint density (or probability) f (evaluated at the data points) as a function θ . Subject to weak technical conditions, it can be shown that the maximum likelihood estimator is consistent and approximately normally distributed in large samples. Specifically, assume that you have n independent observations from some distribution $f(data_i; \theta_0)$, where f is the density, “ $data_i$ ” stands for the data for individual i , and θ_0 is some unknown parameter, which we want to estimate. Then the maximum likelihood estimator maximizes the likelihood function,

$$\mathcal{L}(\theta) \stackrel{def}{=} f(data_1; \theta) \cdot f(data_2; \theta) \cdot \dots \cdot f(data_n; \theta) = \prod_{i=1}^n f(data_i; \theta)$$

Of course the value of θ that maximizes $\mathcal{L}(\theta)$ is the same as the value that

maximizes

$$\log \mathcal{L}(\theta) = \sum_{i=1}^n \log f(\text{data}_i; \theta)$$

which is easier to work with (if you ever need to program any of this yourselves). It can be shown that the maximum likelihood estimator satisfies

$$\hat{\theta}_{MLE} \overset{\text{approx.}}{\sim} N\left(\theta_0, \left(-\frac{\partial^2 \log \mathcal{L}(\theta)}{\partial \theta \partial \theta'}\right)^{-1}\right) \quad (12)$$

where “ $\overset{\text{approx.}}{\sim}$ ” means that the result is approximately true in large samples. It can also be shown that in large samples, the maximum likelihood estimator has the lowest variance among all unbiased estimators.

Remark 4 *All of this can be made formally correct, but you should keep in mind that to do so, we would need to define more precisely what we mean by “ $\overset{\text{approx.}}{\sim}$ ”. We would also need to discuss some technical assumptions that must be made.*

Remark 5 *You are not responsible for the paragraph above on the final for this course.*

9 Panel Data.

Panel data sets have more than one observation for each unit. For example an individual, a firm, a state, a country, etc. may be observed for more than one time-period.

Panel data is also called *longitudinal data*, or *time-series cross-section data*.

Here, we will be interested in the linear panel data model

$$y_{it} = \alpha_i + x'_{it}\beta + \varepsilon_{it}, \quad t = 1, \dots, T_i \quad i = 1, \dots, n.$$

When $T_i = T$ for all i , we say that the panel is balanced. Otherwise, it is unbalanced.

Throughout, we will use the notation

$$y_{i\bullet} = \frac{1}{T_i} \sum_{t=1}^{T_i} y_{it}, \quad x_{i\bullet} = \frac{1}{T_i} \sum_{t=1}^{T_i} x_{it}, \quad \varepsilon_{i\bullet} = \frac{1}{T_i} \sum_{t=1}^{T_i} \varepsilon_{it}.$$

Note that

$$y_{i\bullet} = \alpha_i + x'_{i\bullet}\beta + \varepsilon_{i\bullet}$$

Panel data methods are used in almost all areas of applied economics. Here are a few examples:

- Panel data methods have been used in labor economics to study wage and hours equations. Here the panel typically consists of a individuals observed over time. But there are other cases as well. For example, one study considers a set of pairs of twins. Using the notation above, i would then denote a pair of twins, while t denotes the individual within the pair.
- Panel data methods have been used in finance to study the dividend and investment decisions by firms. Here the panel consists of firms observed over time.
- Panel data methods have been used in international macro to study the relationship between investment and output, profits and cost of capital using yearly of data for different regions of Europe.
- Panel data methods have been used in health economics to study the relationship between pollution infant mortality.
- Panel data methods have been used in macro to study whether or not somebody is “participating” in the stock market.
- Panel data methods have been used in industrial organization to study productivity of firms.
- etc., etc., etc.,...

9.1 Fixed Effects.

There are two basic ways to think about the person-specific intercept, α_i . The random effects approach assumes that α_i can be treated as an error-term.

The fixed effects approach makes no assumptions regarding α_i , except that it is constant over time. In other word you can think of α_i as an individual specific constant term (in which case it is a number) or as a random variable which may depend on the x_{it} in an arbitrary way.

The fixed effects approach eliminated the effect of α_i by considering

$$(y_{it} - y_{i\bullet}) = (x_{it} - x_{i\bullet})' \beta + (\varepsilon_{it} - \varepsilon_{i\bullet})$$

where we are implicitly assuming that x_{it} does not include an intercept. If

$$E[(\varepsilon_{it} - \varepsilon_{i\bullet}) | (x_{it} - x_{i\bullet})] = 0,$$

then OLS estimation will yield estimates of β . that have all the usual nice properties.

It is possible to show that OLS on the deviations from means is numerically identical to the coefficients of x_{it} in the regression of y_{it} on $d_1, d_2, \dots, d_n$ and x_{it} , where $d_i = 1$ for the i 'th individual and 0 otherwise.

9.2 Random Effects.

The random effects specification treats α_i as an error term. It will be assumed that $E[\alpha_i] = 0$, $V[\alpha_i] = \sigma_\alpha^2$, $E[\varepsilon_{it}] = 0$, $V[\varepsilon_{it}] = \sigma_\varepsilon^2$, $cov[\alpha_i, \varepsilon_{it}] = 0$ and $cov[\varepsilon_{is}, \varepsilon_{it}] = 0$ ($s \neq t$) (in addition, we will also assume that $(\alpha_i, \varepsilon_{i1}, \dots, \varepsilon_{iT_i})$ is uncorrelated with $(x_{i1}, \dots, x_{iT_i})$). In the random effects setting the x 's may contain an intercept.

9.3 Some Terminology.

Finally you should know the following terminology: The estimator based on deviations from means is called the *within estimator*, because it exploits the variation within the data for individual i . The *between estimator* exploits the variation between different individuals by estimating the model:

$$y_{i\bullet} = x'_{i\bullet} \beta + \alpha_i + \varepsilon_{i\bullet}, \quad i = 1, \dots, n.$$

Remark 6 *As I am sure you suspect, I could spend a l-o-n-g time on this topic...*